

DANDI:001333: Parkinson's Electrophysiological Signal Dataset (PESD)

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in `dandi_export/` (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from `sub-parkinson-simulated-lfp_ses-111_ecephys.nwb`.

Field	Value (DANDI metadata)
Name	Parkinson's Electrophysiological Signal Dataset (PESD)
Identifier / version	DANDI:001333 / 0.250327.2220
Citation	Biswas, Ananna (2025) Parkinson's Electrophysiological Signal Dataset (PESD) (Version 0.250327.2220) [Data set]. DANDI Archive. https://doi.org/10.48324/dandi.001333/0.250327.2220
DOI	10.48324/dandi.001333/0.250327.2220
Contributors (roles)	Biswas, Ananna (ContactPerson)
License	CC-BY-4.0
Files / size / subjects	52 files, 0.03 GB, 5 subjects
Species	Homo sapiens - Human
Approach	electrophysiological approach
Measurement technique	analytical technique; signal filtering technique; surgical technique; multi electrode extracellular electrophysiology recording technique
Variables measured	ElectricalSeries, LFP, ElectrodeGroup, ProcessingModule
Date published	2025-03-27T22:20:53.652589+00:00

Description (DANDI metadata, verbatim)

The dataset contains electrophysiological signals from both healthy and parkinsonian subjects. We generated two types of samples from each group. The parkinsonian signals show a relatively high power density at the beta frequency (13 to 30 Hz). Thus, the beta oscillations (13 to 30 Hz) in the subthalamic nucleus (STN) are typically used as the pathological biomarkers for PD symptoms. Each sample includes two types of signals: Beta Average Rectified Voltage (ARV) and Local Field Potential (LFP) from the Subthalamic Nucleus (STN). The ARV signals are in the frequency domain and LFP signals are in the time domain.

Beta ARV Signal: The controller beta values are determined by calculating the Average Rectified Value (ARV) of the beta band. This is achieved by fully rectifying the filtered LFP signal using a fourth-order Chebyshev band-pass filter with an 8 Hz bandwidth, centered around the peak of the LFP power spectrum. Local Field Potential (LFP) - STN: Local Field Potentials are derived from the synchronized activity of neuron populations between the cortex, STN, and thalamus.

More details can be found in our article named, "Preliminary Results of Neuromorphic Controller Design and a Parkinson's Disease Dataset Building for Closed-Loop Deep Brain Stimulation", available at <https://arxiv.org/abs/2407.17756>

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
session_description	Parkinson's Electrophysiological Signal Dataset (PESD) Generated from Simulation
experiment_description	The PESD dataset is generated from a cortico-basal-ganglia network for a Parkinsonian computational model. The computational model of the cortico-basal-ganglia is originally presented by Fleming et al. in the article: 'Simulation of Closed-Loop Deep Brain Stimulation Control Schemes for Suppression of Pathological Beta Oscillations in Parkinson's Disease'.
institution	Michigan Technological University
lab	BrainX Lab
experimenter	Ananna Biswas
keywords	ecephys; LFP; Parkinson's Disease; Beta Band

Field	Value (NWB file)
related_publications	https://arxiv.org/abs/2407.17756 ; DOI: 10.3389/fnins.2020.00166
subject	subject_id: parkinson-simulated-lfp; species: Homo sapiens; sex: U; age: P0D; description: This is a simulated dataset generated from a computational model.
devices	NEURON_Simulator: Virtual probe used in NEURON simulation
electrode_groups	shank0: description = 'Simulated electrode group for shank 0', location = 'Simulated Cortico-basal-ganglia network of brain'; shank1: description = 'Simulated electrode group for shank 1', location = 'Simulated Cortico-basal-ganglia network of brain'; shank2: description = 'Simulated electrode group for shank 2', location = 'Simulated Cortico-basal-ganglia network of brain'; shank3: description = 'Simulated electrode group for shank 3', location = 'Simulated Cortico-basal-ganglia network of brain'

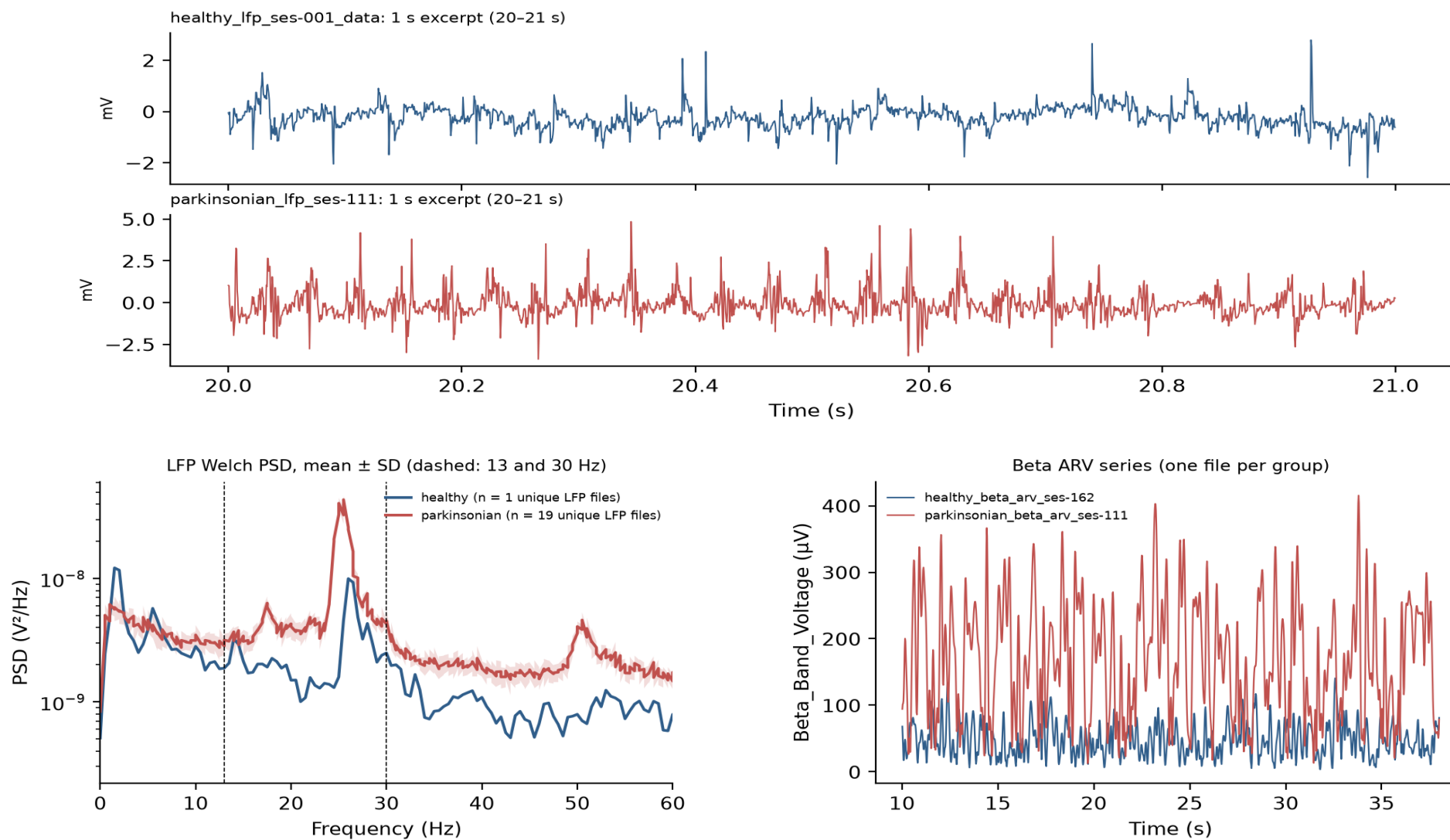
2. Contents of the files

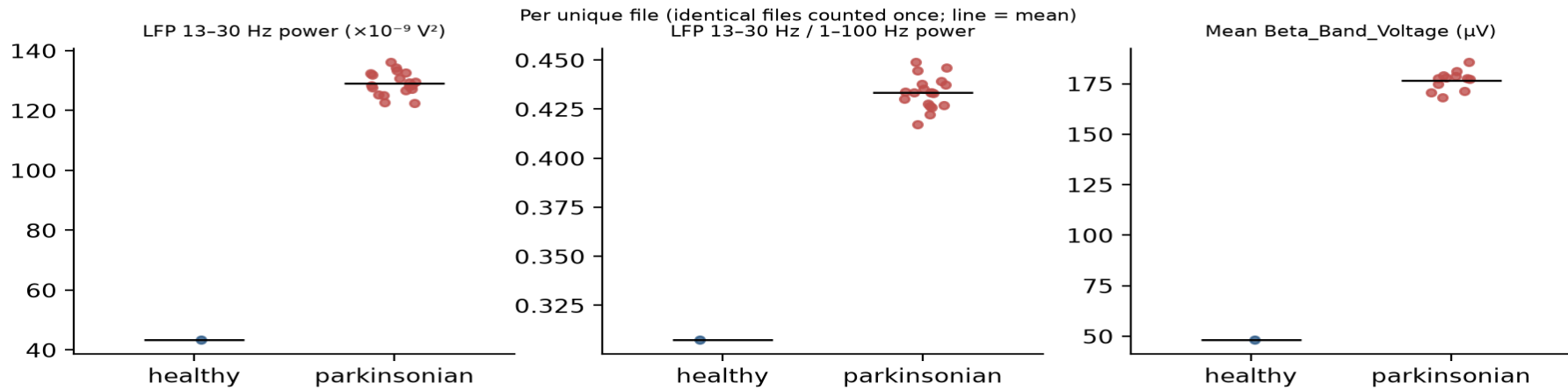
Quantity	Value	Source
Files per NWB subject_id	healthy-simulated-beta (10), healthy-simulated-data (1), healthy-simulated-lfp (10), parkinson-simulated-beta (12), parkinson-simulated-lfp (19)	NWB file
subject.description (all files)	This is a simulated dataset generated from a computational model.	NWB file
subject species / sex / age (all files)	Homo sapiens / U / P0D	NWB file
Device	NEURON_Simulator: Virtual probe used in NEURON simulation	NWB file
Electrode groups	4 groups; location 'Simulated Cortico-basal-ganglia network of brain'	NWB file
LFP series ('LFP')	30 files; 2000 Hz; durations 38.7 s (23), 32.75 s (7)	NWB file
Beta ARV series ('Beta_Band_Voltage')	22 files; timestamps every 0.02 s from 10.02 to 38.00 s (1400 samples)	NWB file
Beta ARV definition	Given in the DANDI description (section 1): 'fully rectifying the filtered LFP signal using a fourth-order Chebyshev band-pass filter with an 8 Hz bandwidth, centered around the peak of the LFP power spectrum'	DANDI metadata
Unique signals (by MD5)	healthy beta_arv: 1 of 10, healthy lfp: 1 of 11, parkinsonian beta_arv: 12 of 12, parkinsonian lfp: 19 of 19	Computed by this export

3. Export parameters

Parameter	Definition	Source
group, signal	from the NWB subject_id: 'healthy'/'parkinson' prefix; '-beta' → beta_arv, '-lfp' or '-data' → lfp	Computed by this export
Power spectrum	Welch, 2 s Hann segments (0.5 Hz resolution), mean removed; 0–100 Hz kept	Computed by this export
beta_power_v2	PSD integrated over 13–30 Hz (the band named in the DANDI description)	Computed by this export
relative_beta_power	beta_power / power over 1–100 Hz	Computed by this export
peak_freq_hz	frequency of the PSD maximum within 5–45 Hz	Computed by this export
unique, duplicate_of	MD5 of each file's signal values; duplicate_of names the first file with identical values	Computed by this export

4. Figures (Computed by this export)





5. Summary per group and signal, unique files (Computed by this export)

Group	Signal	Unique files	13-30 Hz power mean (V^2)	SD	Relative power mean	SD	Peak frequencies (Hz)	Signal SD mean (V)	Signal mean (V)
healthy	beta_arv	1						2.469e-05	4.814e-05
healthy	lfp	1	4.333e-08		0.307		26	5.429e-04	-2.306e-04
parkinsonian	beta_arv	12						8.975e-05	1.767e-04
parkinsonian	lfp	19	1.291e-07	3.81e-09	0.433	0.008	25, 25.5, 25.5	8.416e-04	-9.347e-05

6. All files

Sample	Group	Signal	Samples	fs (Hz)	Duration (s)	Unique	Identical to	SD (V)	13–30 Hz power (V ²)	Relative	Peak (Hz)
healthy_beta_arv_ses-162	healthy	beta_arv	1400	50	28.00	yes		2.469e-05			
healthy_beta_arv_ses-218	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-279	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-345	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-423	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-452	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-516	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-54	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-646	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
healthy_beta_arv_ses-78	healthy	beta_arv	1400	50	28.00	no	healthy_beta_arv_ses-162	2.469e-05			
parkinsonian_beta_arv_ses-111	parkinsonian	beta_arv	1400	50	28.00	yes		8.493e-05			
parkinsonian_beta_arv_ses-112	parkinsonian	beta_arv	1400	50	28.00	yes		9.117e-05			
parkinsonian_beta_arv_ses-120	parkinsonian	beta_arv	1400	50	28.00	yes		8.874e-05			
parkinsonian_beta_arv_ses-163	parkinsonian	beta_arv	1400	50	28.00	yes		9.226e-05			
parkinsonian_beta_arv_ses-164	parkinsonian	beta_arv	1400	50	28.00	yes		8.848e-05			
parkinsonian_beta_arv_ses-284	parkinsonian	beta_arv	1400	50	28.00	yes		9.005e-05			
parkinsonian_beta_arv_ses-289	parkinsonian	beta_arv	1400	50	28.00	yes		9.216e-05			
parkinsonian_beta_arv_ses-298	parkinsonian	beta_arv	1400	50	28.00	yes		8.858e-05			
parkinsonian_beta_arv_ses-339	parkinsonian	beta_arv	1400	50	28.00	yes		8.851e-05			
parkinsonian_beta_arv_ses-51	parkinsonian	beta_arv	1400	50	28.00	yes		8.864e-05			
parkinsonian_beta_arv_ses-60	parkinsonian	beta_arv	1400	50	28.00	yes		8.987e-05			
parkinsonian_beta_arv_ses-93	parkinsonian	beta_arv	1400	50	28.00	yes		9.362e-05			
healthy_lfp_ses-001_data	healthy	lfp	77400	2000	38.70	yes		5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-162	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-218	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-279	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-345	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-423	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-452	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-516	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-54	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-646	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
healthy_lfp_ses-78	healthy	lfp	77400	2000	38.70	no	healthy_lfp_ses-001_data	5.429e-04	4.333e-08	0.307	26.0
parkinsonian_lfp_ses-111	parkinsonian	lfp	77400	2000	38.70	yes		8.367e-04	1.249e-07	0.430	25.0
parkinsonian_lfp_ses-112	parkinsonian	lfp	77400	2000	38.70	yes		8.397e-04	1.276e-07	0.434	25.5
parkinsonian_lfp_ses-120	parkinsonian	lfp	77400	2000	38.70	yes		8.450e-04	1.284e-07	0.422	25.0
parkinsonian_lfp_ses-124	parkinsonian	lfp	65500	2000	32.75	yes		8.477e-04	1.293e-07	0.428	25.5
parkinsonian_lfp_ses-159	parkinsonian	lfp	65500	2000	32.75	yes		8.361e-04	1.224e-07	0.417	25.5
parkinsonian_lfp_ses-163	parkinsonian	lfp	77400	2000	38.70	yes		8.421e-04	1.308e-07	0.446	25.0
parkinsonian_lfp_ses-164	parkinsonian	lfp	77400	2000	38.70	yes		8.315e-04	1.267e-07	0.438	25.0

Sample	Group	Signal	Samples	fs (Hz)	Duration (s)	Unique	Identical to	SD (V)	13–30 Hz power (V ²)	Relative	Peak (Hz)
parkinsonian_lfp_ses-219	parkinsonian	lfp	65500	2000	32.75	yes		8.521e-04	1.334e-07	0.426	25.5
parkinsonian_lfp_ses-245	parkinsonian	lfp	65500	2000	32.75	yes		8.491e-04	1.295e-07	0.427	25.5
parkinsonian_lfp_ses-284	parkinsonian	lfp	77400	2000	38.70	yes		8.426e-04	1.282e-07	0.433	25.0
parkinsonian_lfp_ses-289	parkinsonian	lfp	77400	2000	38.70	yes		8.392e-04	1.323e-07	0.445	25.5
parkinsonian_lfp_ses-298	parkinsonian	lfp	77400	2000	38.70	yes		8.432e-04	1.271e-07	0.434	25.0
parkinsonian_lfp_ses-318	parkinsonian	lfp	65500	2000	32.75	yes		8.412e-04	1.320e-07	0.433	25.5
parkinsonian_lfp_ses-339	parkinsonian	lfp	77400	2000	38.70	yes		8.464e-04	1.327e-07	0.435	25.5
parkinsonian_lfp_ses-51	parkinsonian	lfp	77400	2000	38.70	yes		8.322e-04	1.252e-07	0.433	25.5
parkinsonian_lfp_ses-60	parkinsonian	lfp	77400	2000	38.70	yes		8.376e-04	1.285e-07	0.438	25.0
parkinsonian_lfp_ses-66	parkinsonian	lfp	65500	2000	32.75	yes		8.453e-04	1.343e-07	0.439	25.5
parkinsonian_lfp_ses-69	parkinsonian	lfp	65500	2000	32.75	yes		8.336e-04	1.228e-07	0.427	25.5
parkinsonian_lfp_ses-93	parkinsonian	lfp	77400	2000	38.70	yes		8.487e-04	1.361e-07	0.449	25.5

Signal values: NWB file. Durations, uniqueness and spectral values: computed by this export.

7. Data checks performed by this export

Identical files. Comparing the MD5 hash of each file's signal values: 33 of 52 files are unique. 19 files are identical to another file: healthy_beta_arv: 9 files identical to healthy_beta_arv_ses-162; healthy_lfp: 10 files identical to healthy_lfp_ses-001_data. Every parkinsonian file is unique.

Beta ARV timing. The Beta_Band_Voltage series has no *rate* field; timing comes from its timestamps (every 0.02 s).

8. Column and signal descriptions from the NWB files

Quoted verbatim from sub-parkinson-simulated-lfp_ses-111_ecephys.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	location	the location of channel within the subject e.g. brain region
electrodes	group	a reference to the ElectrodeGroup this electrode is a part of
electrodes	group_name	the name of the ElectrodeGroup this electrode is a part of
electrodes	label	label of electrode
processing/ecephys	(module)	Processed electrophysiology data
processing/ecephys/LFP	LFP	no description

9. Exported files

File	Rows	Columns
tables/sessions.csv	52	sample_id, file, group, signal, session, nwb_series, n_samples, fs_hz, t_start_s, t_end_s, duration_s, md5, duplicate_of, unique, mean_v, sd_v, rms_v, min_v, max_v, beta_power_v2, total_power_1_100_v2, relative_beta_power, peak_freq_hz, peak_psd_v2_per_hz
tables/psd.csv	6,000	sample_id, group, unique, freq_hz, psd_v2_per_hz (LFP files, 0–100 Hz)
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		the published metadata quoted in sections 1 and 8
signals/<sample_id>.csv	one per NWB file	time_s, voltage_v